Agentic AI Assistance for Content Creators

A Project Report

submitted by

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in partial fulfilment of requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY



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I, **BHARGAVA TEJA N**, with Roll No: **ME22B2006** hereby declare that the material presented in the Project Report titled **Agentic AI Assistance for Content Creators** represents original work carried out by me in the **Department of Mechanical Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Date: 11.05.2026

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ABSTRACT

In today's digital-first professional environment, content creation has become an essential part of communication, branding, and outreach. However, content production is not merely about writing text; it involves research, validation of facts, structuring ideas, drafting scripts, refining tone, and ensuring contextual accuracy. For busy professionals, especially those operating in dynamic fields such as law, finance, or corporate governance, this process consumes considerable time and cognitive effort.

This project presents the design and implementation of an Agentic AI-based content research and script generation system built using n8n as an orchestration framework, integrated with real-time retrieval tools (SERP API, Tavily) and Retrieval-Augmented Generation (RAG) using Large Language Models (LLMs).

The system enables a professional user to trigger workflows via WhatsApp or Telegram bots. It retrieves the latest domain-specific news articles, stores them in structured format (Google Sheets), and uses them as grounded references for script generation. A Human-in-the-Loop approval mechanism ensures that the generated content is validated and refined iteratively before acceptance. Beyond script generation, the extended architecture includes advanced media automation capabilities such as voice cloning using ElevenLabs, avatar cloning for personalized video presence, AI-based background image generation aligned with content themes, and automated video composition including intro and outro integration through media processing tools. Once approved, the system is designed to support seamless publishing across multiple social media platforms, including YouTube, Instagram, LinkedIn, Facebook, and Telegram, thereby eliminating repetitive manual uploads.

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CHAPTER 1

INTRODUCTION

1.1 Background

TIn recent years, digital platforms have evolved into primary communication channels for professionals across domains such as law, education, corporate leadership, consulting, and manufacturing. Establishing an online presence is no longer optional; it has become an essential part of knowledge sharing, branding, and stakeholder engagement. Many professionals now regularly publish video content explaining current events, policy updates, domain-specific insights, or organizational announcements.

However, producing digital content is not limited to drafting a script. The process typically includes identifying relevant topics, researching the latest developments, verifying sources, structuring a coherent narrative, recording video, editing visuals, generating suitable background elements, adding branding components such as intro and outro segments, and finally publishing the content across multiple social media platforms. Each of these stages demands both time and technical effort.

While Large Language Models (LLMs) have simplified text generation, they do not independently handle real-time information retrieval, structured workflow control, media processing, or distribution. Furthermore, standalone AI tools lack integration with voice cloning systems, avatar-based video generation platforms, and multi-platform publishing pipelines. As a result, professionals often depend on fragmented tools and manual processes, which reduce efficiency and increase cognitive workload.

1.2 Problem Statement

Despite the availability of advanced AI models and automation platforms, there is currently no unified system that provides an end-to-end solution for professional content creation and distribution. Professionals face several recurring challenges:

1. Time-consuming manual research to stay updated with the latest domain-specific developments.
2. Risk of factual inaccuracies when using generic AI text generators without grounding.
3. Repetitive drafting and redrafting of scripts to achieve the desired tone and personalization.
4. Dependence on separate tools for video recording, editing, and branding elements such as intro and outro segments.
5. Manual effort required to generate suitable background visuals aligned with the content theme.
6. Additional complexity in integrating voice cloning and avatar-based video generation for a consistent digital presence.
7. Repetitive uploading of the same content across multiple social media platforms, including YouTube, Instagram, LinkedIn, Facebook, and Telegram.

These fragmented workflows result in inefficiency, reduced productivity, and increased mental fatigue. Without structured approval loops, personalization and professional authenticity may also be compromised.

Therefore, there is a need for an integrated, agent-based system that can:

- Retrieve real-time information from reliable sources.
- Ground script generation using Retrieval-Augmented Generation (RAG).
- Enable iterative human validation and refinement.
- Support voice cloning and avatar-based video production.
- Automate background image generation and video composition.
- Seamlessly publish finalized content across multiple platforms.

The absence of such an orchestrated system forms the core problem addressed in this project.

CHAPTER 2

MOTIVATION

2.1 Observed Real-World Inefficiencies

During interactions with professionals who actively produce content, several recurring patterns were observed:

- Research takes longer than writing.
- Repeated searching across different websites.
- Copy–paste workflow between multiple tabs.
- Difficulty maintaining a structured content flow.

The overall process is mentally exhausting and inefficient.

2.2 Limitations of Direct LLM Usage

While Large Language Models (LLMs) can generate content quickly, they exhibit critical drawbacks:

- Reliance on parametric knowledge.
- Possibility of producing outdated or fabricated facts.
- Inability to verify real-time developments.
- Lack of structured workflow governance.

Using LLMs without retrieval grounding increases the risk of misinformation.

2.3 Why Agentic Architecture?

An agentic system differs significantly from a simple chatbot. An agent:

- Receives a defined goal.
- Collects and utilizes necessary tools.

- Retrieves relevant information.
- Generates structured output.
- Waits for user validation.
- Iterates based on feedback.

This structured decision-making pipeline motivated the architecture of the proposed system.

CHAPTER 3

LITERATURE SURVEY

3.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a hybrid AI architecture that combines:

- Dense retrieval systems
- Generative language models

Instead of depending solely on pretrained knowledge, the model retrieves relevant documents at runtime and conditions its generation on the retrieved content.

Advantages:

- Reduces hallucinations
- Ensures domain relevance
- Allows real-time adaptability

This project applies RAG principles by grounding script generation in retrieved news articles stored in Google Sheets.

3.2 Workflow Orchestration Systems

Automation frameworks such as *n8n* provide:

- Node-based workflow design
- API integration capabilities
- Conditional branching
- Error handling

Unlike traditional scripts, n8n enables visual, modular, and scalable workflow orchestration, making it ideal for multi-stage AI pipelines.

3.3 Human-in-the-Loop AI Systems

Modern AI research emphasizes maintaining human involvement in decision-making pipelines.

Benefits include:

- Quality control
- Ethical oversight
- Personalization
- Accountability

This project incorporates an explicit approval and regeneration loop, aligning it more closely with enterprise-grade AI systems rather than standalone chatbot tools.

CHAPTER 4

OBJECTIVES

The current objectives focuses on implementing the foundational components of the proposed Agentic AI-based content research and script generation system. The objectives are outlined as follows:

1. **To implement workflow triggering through WhatsApp and Telegram interfaces.** This involves integrating WhatsApp using Green API and Telegram using BotFather to enable real-time, webhook-based workflow activation through user messages. The goal is to establish a seamless and accessible interaction layer for initiating the automation pipeline.
2. **To enable real-time retrieval of latest news articles using SERP API and Tavily.** The system integrates external retrieval tools to fetch current and domain-relevant news articles. Retrieved data includes titles, URLs, and snippets, ensuring that subsequent content generation is grounded in up-to-date information.
3. **To implement structured storage of retrieved data in Google Sheets.** Retrieved articles are stored in a structured format within Google Sheets, which acts as an external knowledge repository. This ensures traceability and supports contextual referencing during script generation.
4. **To implement Retrieval-Augmented Generation (RAG) for script creation.** The system conditions the Large Language Model on retrieved news content rather than relying solely on pretrained knowledge. This RAG-based approach improves factual accuracy, contextual grounding, and domain relevance.
5. **To develop an iterative human validation and regeneration loop.** Generated scripts are sent to the user for approval. If modifications are required, user feedback is incorporated into a regeneration cycle. This ensures personalization, control, and alignment with professional intent.
6. **To automate AI-driven video generation and media composition.** The approved script is converted into a complete video through multiple AI-driven stages including avatar creation using HeyGen, voice cloning using ElevenLabs, AI-based background generation, and FFmpeg-based editing and merging of intro, outro, transitions, and media layers. This enables fully automated and personalized video production without manual editing.
7. **To enable multi-platform publishing and centralized system monitoring.** The finalized video is automatically published across platforms such as YouTube, Instagram, Facebook, LinkedIn, and Telegram using platform APIs. Additionally,

the system maintains centralized logging of workflow execution, generated content, uploaded URLs, and monitoring data, while also sending real-time error alerts to the backend team for rapid issue resolution and production-ready reliability.

These objectives establish the intelligence, retrieval, and validation layers of the system, forming the core foundation for future extensions such as automated video generation and multi-platform publishing.

CHAPTER 5

METHODOLOGY

5.1 Architectural Design

5.1.1 Architectural Philosophy

The system was designed based on the following core principles:

- Modularity
- Transparency
- Controlled generation
- Iterative refinement
- Scalable integration

5.1.2 High-Level Architecture

The architecture consists of six major layers:

1. **Interaction Layer**
    - WhatsApp Bot (Green API integration)
    - Telegram Bot (BotFather configuration)
2. **Orchestration Layer**
    - n8n workflow engine
    - Conditional nodes
    - Data routing logic
3. **Retrieval Layer**
    - SERP API integration
    - Tavily API integration
    - Article extraction

4. **Storage Layer**
    - Google Sheets as structured datastore
5. **Intelligence Layer**
    - LLM-based script generation
    - RAG context conditioning
6. **Validation Layer**
    - Approval
    - Feedback capture
    - Regeneration agent
7. **Media Layer**
    - Voice Cloning
    - Avatar Generation
    - Background Generation
    - Video Composition, Editing
    - Video Approval Loop
8. **Social Media Publishing Layer**
    - Metadata Generation
    - Multi-Platform Upload
9. **Monitoring Layer**
    - Logging, Monitoring
    - Error Handling

5.1.3 Detailed Workflow Explanation

Step 1: Workflow Trigger

The user sends a message via:

- WhatsApp (Green API integration)
- Telegram (BotFather-generated bot token)

The n8n webhook captures this input.

Step 2: News Retrieval

Using:

- SERP API
- Tavily

The system retrieves:

- Latest news articles
- Title
- URL
- Snippet

These results are parsed and structured.

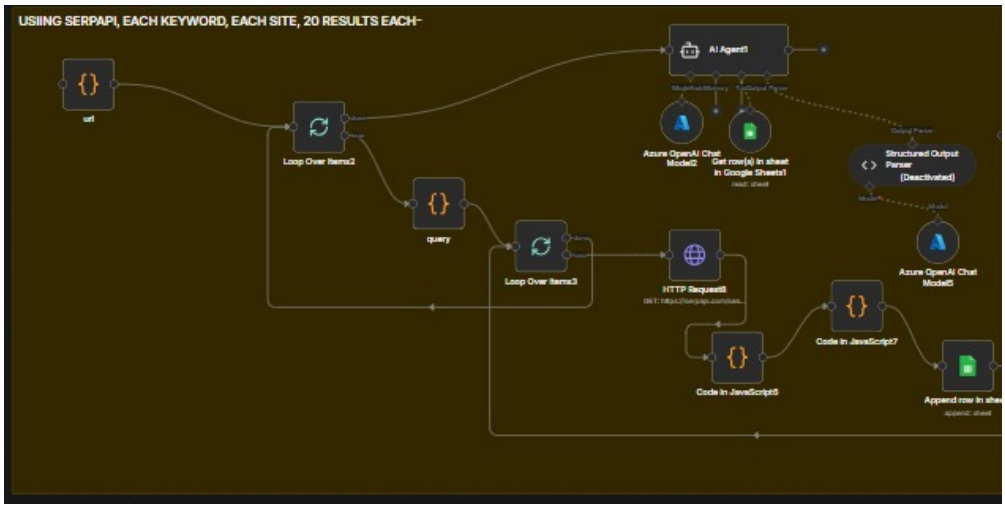


Figure 5.1: News retrieval workflow overview.

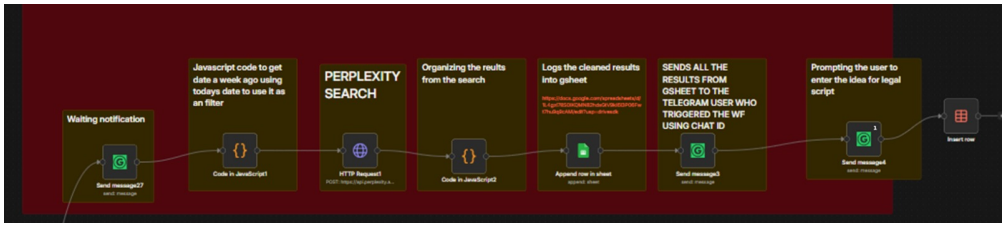


Figure 5.2: Storing the results in data base

Step 3: Data Storage

All retrieved articles are:

- Cleaned
- Structured
- Stored in Google Sheets

This acts as a lightweight database and reference memory.

Step 4: RAG-Based Script Generation

This is the core intelligence stage.

The process includes:

- Extracting relevant news from Google Sheets
- Constructing a structured prompt
- Injecting retrieved content as context
- Conditioning the LLM to rely strictly on this context

The model generates a structured script grounded in retrieved information.

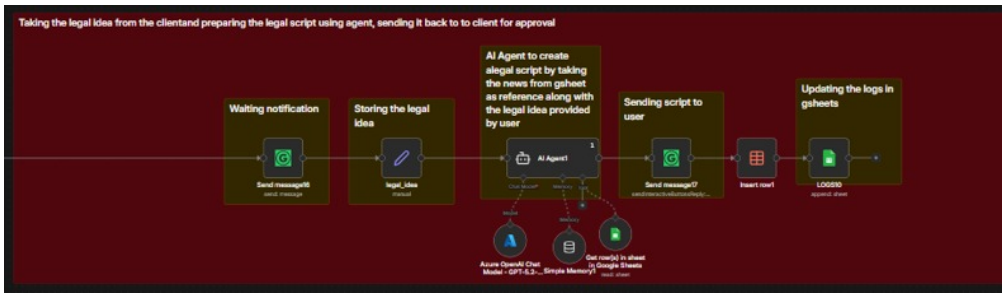


Figure 5.3: RAG-based script generation workflow.

Step 5: Human Approval

The process includes:

- Sending the generated script to the user through WhatsApp or Telegram

- Allowing the user to approve, reject, or modify the generated content
- Collecting user feedback for refinement and personalization
- Ensuring human oversight before proceeding to media generation

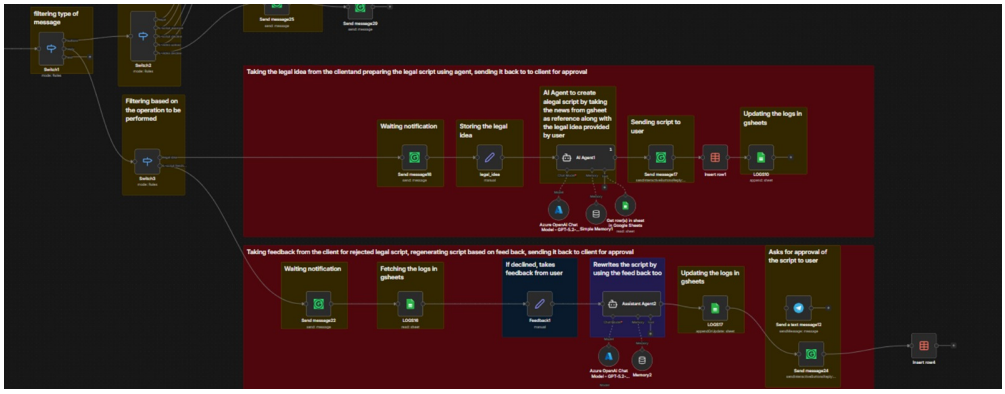


Figure 5.4: Human-in-the-loop approval flow.

Step 6: Voice Cloning (ElevenLabs)

The process includes:

- Converting the approved script into realistic speech using ElevenLabs API
- Utilizing the client's pre-trained voice model for personalized narration
- Preserving natural tone, pacing, and speaking style
- Eliminating the need for manual voice recording

Step 7: Avatar Generation (HeyGen)

The process includes:

- Generating an AI avatar synchronized with the cloned audio
- Performing automatic lip-sync and facial expression alignment
- Creating realistic upper-body presentation videos
- Producing professional video content without manual recording

Step 8: Background Generation

The process includes:

- Analyzing script context to generate relevant visual backgrounds
- Creating AI-generated images or short visual loops
- Improving thematic consistency and audience engagement
- Eliminating manual background selection and editing

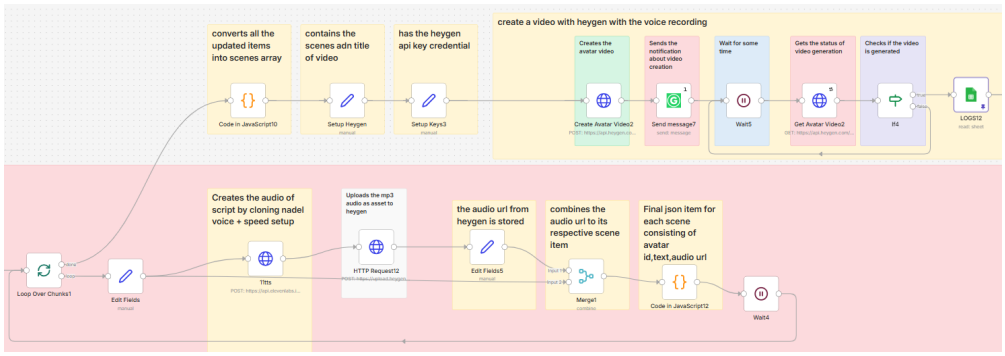


Figure 5.5: Video Generation using heygen,11labs

Step 9: Video Composition (FFmpeg)

The process includes:

- Combining avatar video, cloned audio, and generated backgrounds
- Integrating intro, outro, transitions, and overlays
- Rendering the final professional-quality video output
- Automating the complete editing and composition workflow

Step 10: Video Approval Loop

The process includes:

- Sending the generated video back to the user for review
- Allowing approval or rejection based on output quality
- Collecting user feedback for regeneration if required

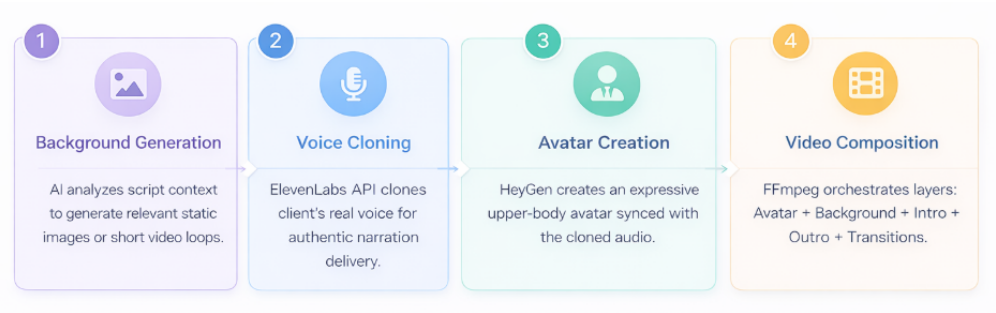


Figure 5.6: Generation pipeline

Step 11: Metadata Generation

The process includes:

- Generating SEO-optimized titles and descriptions using AI
- Creating platform-specific hashtags and tags
- Generating call-to-action content for audience engagement
- Improving discoverability across social media platforms

Step 12: Multi-Platform Upload

The process includes:

- Uploading finalized videos through platform APIs
- Supporting YouTube, Instagram, LinkedIn, Facebook, and Telegram
- Ensuring consistent cross-platform content distribution

Step 13: Logging and Monitoring

The process includes:

- Recording workflow execution details in Google Sheets
- Tracking uploaded URLs, timestamps, and workflow status
- Maintaining centralized monitoring and audit history
- Supporting performance analysis and operational transparency

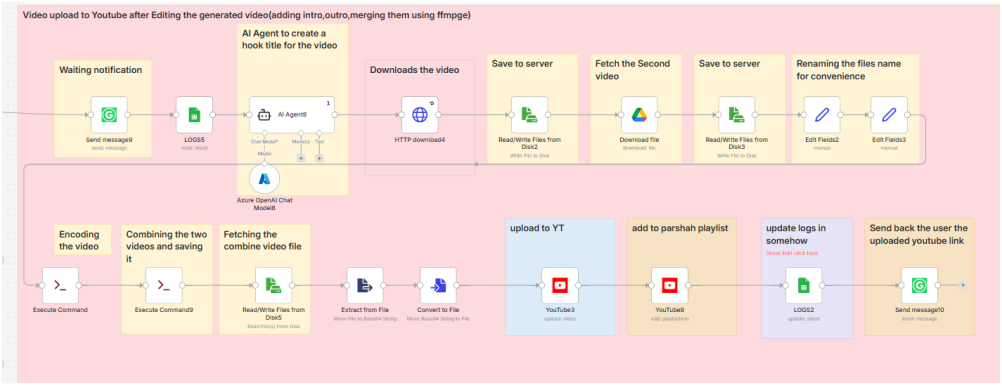


Figure 5.7: Editing,Video publishing, Logging the data

Step 14: Error Handling

The process includes:

- Monitoring APIs and workflow stages continuously
- Detecting failures such as timeouts or authentication errors
- Sending instant alerts through WhatsApp, Telegram, and email
- Enabling rapid debugging and improving system reliability

5.1.4 How RAG Is Implemented in This Project

Unlike theoretical RAG systems that rely on vector databases, this implementation uses a structured datastore (Google Sheets) as the retrieval memory.

Process:

- News articles are treated as authoritative context.
- Relevant articles are selected.
- Context is concatenated into the prompt.
- The LLM is instructed to generate content strictly from the provided material.

This approach prevents hallucination and ensures factual grounding.

CHAPTER 6

IMPLEMENTATION DETAILS

6.1 System Integrations and Workflow Engineering

This chapter presents the technical implementation of the proposed Agentic AI system. The implementation focuses on integrating messaging platforms, external retrieval APIs, structured data storage, and workflow orchestration using n8n. The objective of this phase was to translate the conceptual architecture into a functional, modular, and scalable pipeline capable of handling real-time interactions and iterative script generation.

6.2 Implementation and Integrations

6.2.1 WhatsApp Integration

WhatsApp integration was implemented using Green API to establish real-time communication between the user and the automation pipeline. The integration uses webhook-based triggering to initiate workflow execution whenever a message is received from the client.

The implementation includes:

- Incoming message capture through webhook endpoints
- Bidirectional communication between user and workflow
- Role-based workflow triggering using custom commands
- Real-time response delivery through WhatsApp messaging

6.2.2 Telegram Integration

Telegram integration was implemented using BotFather-generated API credentials. The Telegram bot acts as an alternative communication interface for triggering and monitoring workflow execution.

The integration includes:

- Command-based workflow triggering
- Webhook configuration within n8n
- Structured interaction through Telegram bot messages
- Instant delivery of generated outputs and notifications

6.2.3 SERP API Integration

SERP API was integrated to retrieve real-time search results and latest domain-specific news articles required for content generation.

The implementation includes:

- Dynamic query-based news retrieval
- Structured JSON response handling
- Extraction of article titles, URLs, and snippets
- Real-time retrieval of up-to-date information

6.2.4 Tavily Integration

Tavily was integrated as an AI-optimized semantic retrieval system to improve contextual relevance during information extraction.

The integration includes:

- Semantic search-based article retrieval
- AI-optimized contextual news extraction
- Enhanced relevance filtering for retrieved content
- Structured retrieval for RAG-based generation

6.2.5 Google Sheets Integration

Google Sheets was used as a centralized lightweight database for storing workflow data, retrieval outputs, logs, and monitoring information.

The integration includes:

- Structured storage of retrieved articles
- Logging of generated scripts and workflow outputs
- Storage of uploaded social media links
- Centralized monitoring and execution tracking

6.2.6 RAG Pipeline Integration

Retrieval-Augmented Generation was implemented to ensure that generated content remains grounded in retrieved contextual information instead of relying solely on pre-trained model knowledge.

The implementation includes:

- Retrieval of contextual information from Google Sheets
- Prompt conditioning using retrieved knowledge
- Context-grounded script generation using LLMs
- Reduction of hallucinations and factual inconsistencies

6.2.7 ElevenLabs Voice Cloning Integration

ElevenLabs API was integrated for AI-based voice cloning and personalized narration generation.

The integration includes:

- Conversion of scripts into realistic speech
- Use of client-specific cloned voice models
- Preservation of tone, pacing, and speaking style
- Automated narration generation without manual recording

6.2.8 HeyGen Avatar Integration

HeyGen API was integrated for avatar-based video generation synchronized with generated narration.

The integration includes:

- AI avatar generation using approved scripts
- Lip-sync and facial synchronization with generated audio
- Upper-body expressive video generation
- Automated presenter-style content creation

6.2.9 FFmpeg Integration

FFmpeg was integrated for media composition and automated video editing.

The implementation includes:

- Merging avatar video and generated backgrounds
- Integration of intro and outro animations
- Automated transition and overlay composition
- Final rendering of professional-quality video outputs

6.2.10 Social Media API Integrations

Multiple platform APIs were integrated for automated publishing and distribution of finalized content.

The implementation includes:

- Automated upload to YouTube, Instagram, LinkedIn, Facebook, and Telegram
- Generation of platform-specific metadata
- Elimination of repetitive manual uploading tasks
- Cross-platform content distribution through a unified workflow

6.2.11 Monitoring and Logging System

A centralized monitoring system was implemented using Google Sheets for execution tracking and operational transparency.

The implementation includes:

- Logging of workflow execution details
- Tracking of uploaded URLs and timestamps
- Monitoring of generation status and execution duration
- Centralized audit and performance tracking

6.2.12 Error Handling and Alert System

An automated error detection and notification system was implemented to improve reliability and minimize workflow failures.

The implementation includes:

- Real-time monitoring of APIs and workflow stages
- Detection of authentication failures and API limits
- Instant alerts through WhatsApp, Telegram, and email
- Rapid debugging support using execution logs

6.2.13 n8n Workflow Orchestration

n8n was used as the central orchestration engine responsible for coordinating all workflow stages and external integrations.

The implementation includes:

- Webhook-based workflow triggering
- Conditional routing and decision-making logic
- Multi-agent orchestration and API coordination
- Automated execution of end-to-end content pipelines

CHAPTER 7

DELIVERABLES

Completed Components

The following components have been successfully implemented:

- Functional bot interface
- Real-time retrieval integration
- RAG-based script generation
- Structured approval loop
- Modular architecture
- Automated Video Genaration (Avatar clone, audio clone,Editing,..)
- Socia Media publishing

Current Milestone Achievements

The system lays a strong foundation for:

- Media automation
- Cross-platform distribution
- Enterprise-level AI deployment

It reflects significant effort in:

- Architecture design
- Multi-API integration
- RAG implementation
- Workflow engineering

Overall, the project demonstrates that Agentic AI can transform content workflows across industries where time efficiency is critical.

CHAPTER 8

FUTURE WORK PLAN

8.1 Future Work

A COMPLETE WEB APPLICATION OR MOBILE APPLICATION THAT RUNS END TO END INSTEAD OF WHATSAPP OR TELEGRAM

Cross-Domain Extension

This architecture is domain-agnostic and can be adapted across industries.

Example: Manufacturing Sector

A CEO could:

- Input a notice topic
- System generates a script
- AI avatar delivers the announcement
- Automatically distribute it internally

Other sectors include:

- Marketing
- Education
- Healthcare
- Corporate communication

Time-sensitive environments can significantly benefit from this system.

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